

Pulse-width modulation

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Pulse-width modulation of a signal or power source involves the modulation of its duty cycle, to either convey information over a communications channel or control the amount of power sent to a load.

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Principle

Pulse-width modulation uses a square wave whose duty cycle is modulated resulting in the variation of the average value of the waveform. If we consider a square waveform $f(t)$ with a low value y_{min} , a high value y_{max} and a duty cycle D (see figure 1), the average value of the waveform is given by:

$$\bar{y} = \frac{1}{T} \int_0^T f(t) dt$$

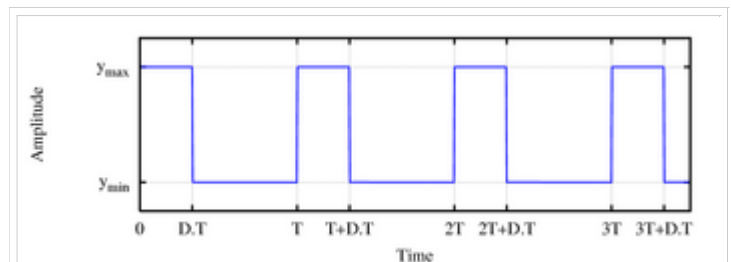


Fig. 1: a square wave, showing the definitions of y_{min} , y_{max} and D .

As $f(t)$ is a square wave, its value is y_{max} for $0 < t < D \cdot T$ and y_{min} for $D \cdot T < t < T$. The above expression then becomes:

$$\begin{aligned} \bar{y} &= \frac{1}{T} \left(\int_0^{DT} y_{max} dt + \int_{DT}^T y_{min} dt \right) \\ &= \frac{D \cdot T \cdot y_{max} + T(1-D)y_{min}}{T} \\ &= D \cdot y_{max} + (1 - D) y_{min} \end{aligned}$$

This latter expression can be fairly simplified in many cases where $y_{min} = 0$ as $\bar{y} = D \cdot y_{max}$. From this, it is obvious that the average value of the signal ($\bar{y}$) is directly dependent on the duty cycle D .

Technique

Generation

Intersective

The simplest way to generate a PWM signal is the intersective method, which requires only a sawtooth or a triangle waveform (easily generated using a simple oscillator) and a comparator. When the value of the reference signal (the green sine wave in figure 2) is more than the modulation waveform (blue), the PWM signal (magenta) is in the high state, otherwise it is in the low state.

Delta

The output signal is compared with limits, which correspond to a reference signal offset by a constant. Every time the output signal reaches one of the limits, the PWM signal changes state.

Sigma-Delta

The output signal is subtracted from a reference signal to form an error signal. This error is integrated, and when the integral of the error exceeds the limits, the output changes state.

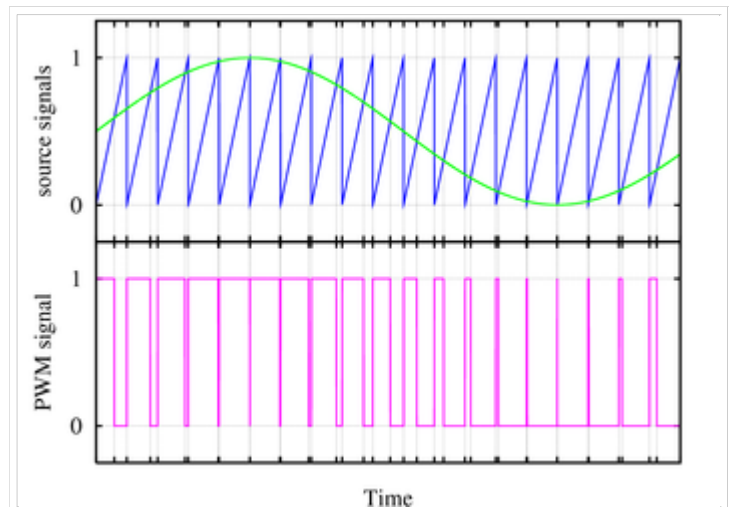


Fig. 2: A simple method to generate the PWM pulse train corresponding to a given signal is the intersective PWM: the signal (here the green sinewave) is compared with a sawtooth waveform (blue). When the latter is less than the former, the PWM signal (magenta) is in high state (1). Otherwise it is in the low state (0).

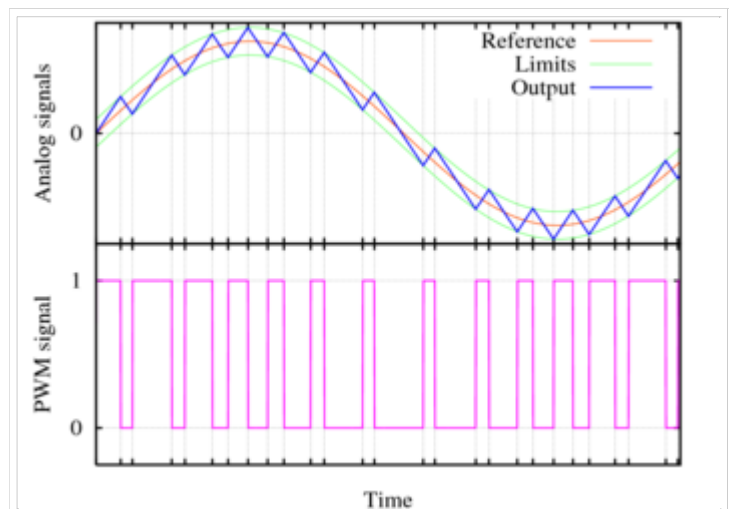


Fig. 3 : Principle of the delta PWM. The output signal (blue) is compared with the limits (gn). These limits correspond to the reference signal (red), offset by a given value. Every time the output signal reaches one of the limits, the PWM signal changes state.

Digital

Many digital circuits can generate PWM signals (e.g many microcontrollers have PWM outputs to control an electrical motor). They usually use a counter that increments periodically (it is connected directly or indirectly to the clock of the circuit) and is reset at the end of every period of the PWM. When the counter value is more than the reference value, the PWM output changes state from high to low.

Types

Three types of pulse-width modulation (PWM) are possible.

1. The pulse center may be fixed in the center of the time window and both edges of the pulse moved to compress or expand the width.
2. The lead edge can be held at the lead edge of the window and the tail edge modulated.
3. The tail edge can be fixed and the lead edge modulated.

Spectrum

The resulting spectra (of the three cases) are similar, and each contains a dc component, a base sideband containing the modulating signal and phase modulated carriers at each harmonic of the frequency of the pulse. The amplitudes of the harmonic groups are restricted by a $\sin x / x$ envelope (sinc function) and extend to infinity.

Applications

Telecommunications

In telecommunications, the widths of the pulses correspond to specific data values encoded at one end and decoded at the other.

Pulses of various lengths (the information itself) will be sent at regular intervals (the carrier frequency of the modulation).

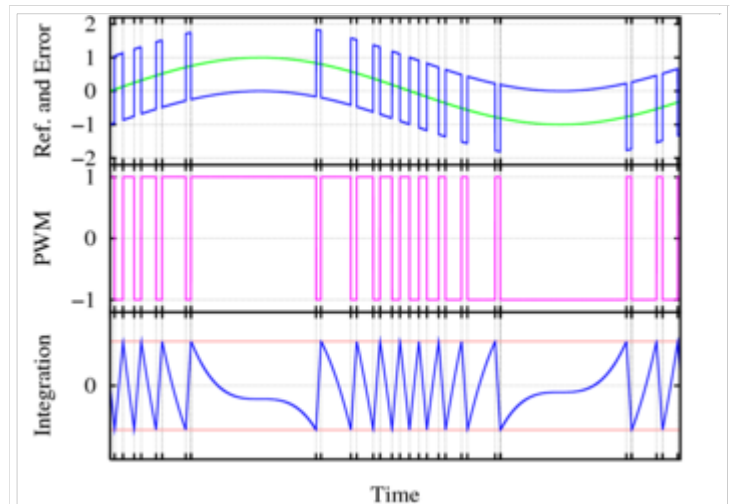
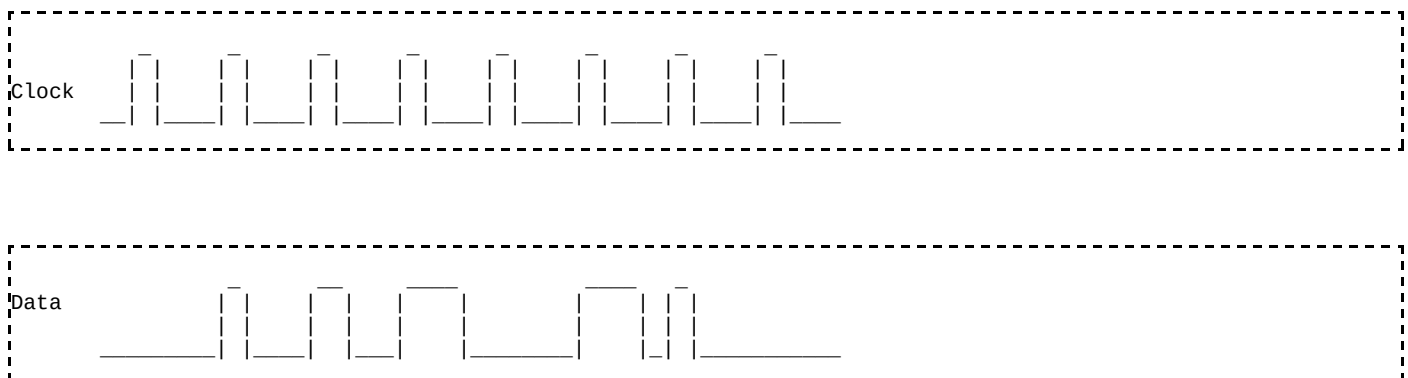


Fig. 4 : Principle of the sigma-delta PWM. The top green waveform is the reference signal, on which the output signal (PWM, in the middle plot) is subtracted to form the error signal (blue, in top plot). This error is integrated (bottom plot), and when the integral of the error exceeds the limits (red lines), the output changes state.

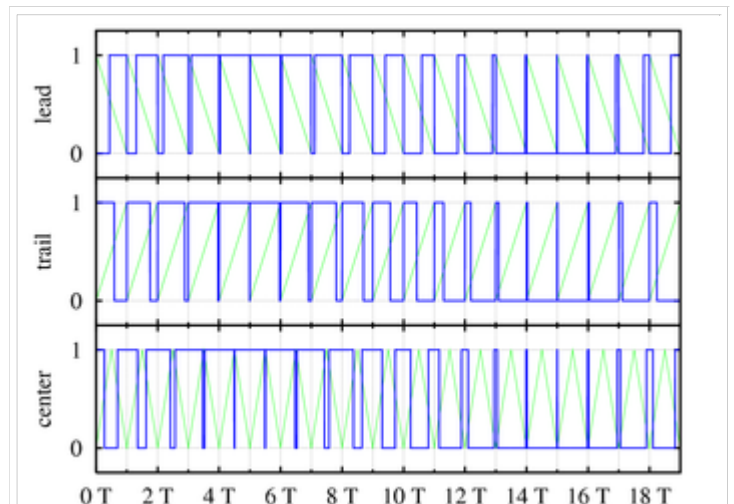


Fig. 5 : Three types of PWM signals (blue): leading edge modulation (top), trailing edge modulation (middle) and centered pulses (both edges are modulated, bottom). The green lines are the sawtooth signals used to generate the PWM waveforms using the interseptive method.

Data	0	1	2	4	0	4	1	0
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The inclusion of a clock signal is not necessary, as the leading edge of the data signal can be used as the clock if a small offset is added to the data value in order to avoid a data value with a zero length pulse.

Power delivery

PWM can be used to reduce the total amount of power delivered to a load without losses normally incurred when a power source is limited by resistive means. This is because the average power delivered is proportional to the modulation duty cycle. With a sufficiently high modulation rate, passive electronic filters can be used to smooth the pulse train and recover an average analog waveform.

High frequency PWM power control systems are easily realisable with semiconductor switches. The discrete on/off states of the modulation are used to control the state of the switch(es) which correspondingly control the voltage across or current through the load. The major advantage of this system is the switches are either off and not conducting any current, or on and have (ideally) no voltage drop across them. The product of the current and the voltage at any given time defines the power dissipated by the switch, thus (ideally) no power is dissipated by the switch. Reallistically, semiconductor switches such as MOSFETs or BJTs are non-ideal switches, but high efficiency controllers can still be built.

PWM is also often used to control the supply of electrical power to another device such as in speed control of electric motors, volume control of Class D audio amplifiers or brightness control of light sources and many other power electronics applications. For example, light dimmers for home use employ a specific type of PWM control. Home use light dimmers typically include electronic circuitry which suppresses current flow during defined portions of each cycle of the AC line voltage. Adjusting the brightness of light emitted by a light source is then merely a matter of setting at what voltage (or phase) in the AC cycle the dimmer begins to provide electrical current to the light source (e.g. by using an electronic switch such as a triac). In this case the PWM duty cycle is defined by the frequency of the AC line voltage (50 Hz or 60 Hz depending on the country). These rather simple types of dimmers can be effectively used with inert (or relatively slow reacting) light sources such as incandescent lamps, for example, for which the additional modulation in supplied electrical energy which is caused by the dimmer causes only negligible additional fluctuations in the emitted light. Some other types of light sources such as light-emitting diodes (LEDs), however, turn on and off extremely rapidly and would perceivably flicker if supplied with low frequency drive voltages. Perceivable flicker effects from such rapid response light sources can be reduced by increasing the PWM frequency. If the light fluctuations are sufficiently rapid, the human visual system can no longer resolve them and the eye perceives the time average intensity without flicker (see flicker fusion threshold).

Voltage regulation

PWM is also used in efficient voltage regulators. By switching voltage to the load with the appropriate duty cycle, the output will approximate a voltage at the desired level. The switching noise is usually filtered with an inductor and a capacitor.

One method measures the output voltage. When it is lower than the desired voltage, it turns on the switch. When the output voltage is above the desired voltage, it turns off the switch.

Audio effects and amplification

PWM is sometimes used in sound synthesis, in particular subtractive synthesis, as it gives a nice effect similar to chorus or slightly detuned oscillators played together. (In fact, PWM is equivalent to the difference of two sawtooth waves. [1] (<http://www.soundonsound.com/sos/Mar03/articles/synthsecrets47.asp>)) The ratio between the high and low level is typically modulated with a low frequency oscillator, or LFO.

A new class of audio amplifiers based on the PWM principle is becoming popular. Called "Class-D amplifiers", these amplifiers produce a PWM equivalent of the analogue input signal which is fed to the loudspeaker via a suitable filter network to block the carrier and recover the original audio. These amplifiers are characterized by very good efficiency figures ($\geq 90\%$) and compact size/light weight for large power outputs.

Historically, a crude form of PWM has been used to play back PCM digital sound on the PC speaker, which is only

capable of outputting two sound levels. By carefully timing the duration of the pulses, and by relying on the speaker's physical filtering properties (limited frequency response, self-inductance, etc.) it was possible to obtain an approximate playback of mono PCM samples, although at a very low quality, and with wildly varying results between implementations.

See also

- Pulse-code modulation
- Pulse-amplitude modulation
- Pulse-density modulation
- Pulse-position modulation
- Radio control
- Switching amplifier

External links

- A PWM circuit using 555 timer ICs (<http://www.eleinmec.com/article.asp?28>)
- Some simple circuits to generate PWM signals (<http://homepages.which.net/~paul.hills/Circuits/PwmGenerators/PwmGenerators.html>)
- Nick's Idea Page :: PWM Modulator (<http://uvasux.googlepages.com/pwmmodulator>) includes the design and discussion of a PWM Modulator circuit used for the control of Servo Motors.
- An Introduction to Delta Sigma Converters (<http://www.beis.de/Elektronik/DeltaSigma/DeltaSigma.html>)
- Fix for 4-pin PWM Fans not running at full speed on legacy motherboards with 3-pin CPU fan headers (<http://www.frostytech.com/permalink.cfm?NewsID=54321>)

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